

Robust Multi-UAV Cooperative Search for Dynamic Targets via Heavy-Tail-Adapted Multi-Agent Reinforcement Learning

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Abstract—Dynamic multi-UAV cooperative search is critical for real-world missions such as disaster response and surveillance, where targets may move unpredictably and detection signals are sparse and noisy. Traditional coverage-based strategies and standard multi-agent reinforcement learning (MARL) approaches struggle in such settings due to outdated spatial beliefs and highly skewed reward distributions. In particular, rare but large detection rewards induce heavy-tailed learning dynamics, leading to unstable policy updates. To address these challenges, we propose a robust hierarchical framework for dynamic multi-UAV search. The system adopts a high–low altitude architecture in which high-altitude UAVs maintain a temporally evolving suspicion heatmap that aggregates probabilistic evidence with adaptive decay, while low-altitude UAVs perform decentralized decision-making guided by this global spatial prior. To improve learning stability under sparse and heavy-tailed rewards, we integrate a heavy-tailed upper confidence bound (HT-UCB) exploration mechanism and a Median-of-Means (MoM) enhanced MAPPO optimizer for robust advantage estimation and policy updates. Extensive simulations demonstrate that the proposed method significantly improves search efficiency and convergence stability in dynamic target scenarios.

I. INTRODUCTION

Cooperative multi-UAV search has become a fundamental approach for large-scale target localization tasks, including disaster response [1], surveillance [2], and urban security [3]. Traditional search strategies typically rely on coverage-based exploration, where UAVs systematically scan predefined regions to locate targets. However, in realistic dynamic scenarios, targets may move over time and prior observations may become unreliable due to occlusion and sensing noise, rendering static coverage assumptions insufficient.

As illustrated in Fig. 1(a), when a target migrates from location A to a new position, previously detected hotspots gradually lose validity. Persisting in outdated high-probability regions results in inefficient motion and wasted search time. Therefore, dynamic search demands not only spatial coverage but also continuous belief updating and adaptive reallocation of search effort.

Recent research on cooperative multi-UAV search can be broadly categorized into rule-based or optimization-driven task allocation methods [4], probabilistic search strategies leveraging belief maps or information gain [5], and multi-agent reinforcement learning (MARL) frameworks under the

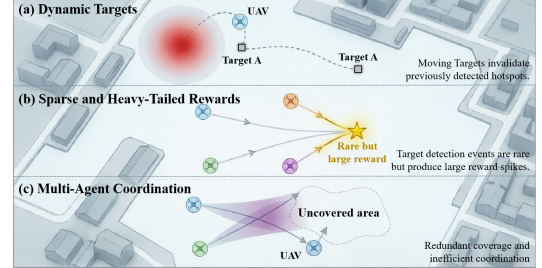


Fig. 1. Challenges in dynamic multi-UAV collaborative search.

centralized training and decentralized execution paradigm. While these approaches have demonstrated effectiveness in static or mildly dynamic settings, they remain limited in fully dynamic cooperative search. The first challenge is the sparse and heavy-tailed nature of reward signals (Fig. 1(b)). Most time steps yield near-zero rewards, whereas successful detections produce rare but large reward spikes. Such distributions amplify variance in advantage estimation and can destabilize policy gradient updates, particularly in multi-agent settings. The second challenge concerns coordination inefficiency (Fig. 1(c)). Without structured coordination, UAVs may redundantly cover the same region while leaving other areas unexplored. Existing hierarchical or scalable MARL approaches primarily focus on policy decomposition or agent scalability, but do not explicitly address temporal belief evolution or robustness under heavy-tailed rewards.

To address these challenges, we propose a robust hierarchical cooperative framework for dynamic multi-UAV search. The framework adopts a heterogeneous high–low altitude architecture: a high-altitude UAV performs large-scale perception and maintains a temporally decaying belief heatmap, enabling adaptive belief reconstruction as targets move. Low-altitude UAVs are guided by this evolving map to conduct localized verification, reducing redundant coverage and improving responsiveness. From the learning perspective, we introduce heavy-tail-aware robust optimization mechanisms. Specifically, a heavy-tailed upper confidence bound (HT-UCB) strategy enhances exploration under statistical uncertainty, while a Median-of-Means (MoM) aggregation is incorporated into MAPPO to mitigate the influence of outliers in advantage estimation, thereby improving training stability in multi-agent settings.

The main contributions are summarized as follows:

- We propose a hierarchical MARL framework that integrates global guidance from high-altitude UAVs with decentralized low-altitude execution.

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- We introduce a temporally evolving suspicion heatmap to fuse noisy detections and enable adaptive spatial belief updating.
- We develop a distributionally robust learning strategy that combines heavy-tailed UCB exploration with a Median-of-Means enhanced MAPPO, improving stability under sparse and heavy-tailed rewards.
- We validate the proposed framework in a realistic 3D simulator, demonstrating strong performance and generalization across varying environments and task scales.

II. RELATED WORKS

Multi-UAV collaborative exploration and collaborative search are closely related yet fundamentally distinct tasks. Exploration emphasizes comprehensive environmental coverage, whereas search prioritizes locating specific targets as quickly as possible. While collaborative exploration has been extensively studied, cooperative search—particularly dynamic target search—remains comparatively underexplored. Based on collaborative decision mechanisms, existing approaches can be broadly categorized into **Multi-Stage Methods** and **Learning-Based Methods** [4].

A. Multi-Stage Methods

Multi-stage methods decompose the search process into target generation, task allocation [6], and motion planning. The environment is typically partitioned using grid-based or Voronoi-based schemes, followed by task allocation based on utility, cost, or expected information gain. Common strategies include iterative optimization [7], [8], metaheuristics [9]–[11], market-based schemes [12], [13], TSP variants [14]–[16], and reinforcement learning-based allocation [17]–[28]. The final stage, motion planning, computes obstacle-avoiding trajectories that satisfy environmental constraints.

These approaches provide structured coordination and interpretability in static or fully observable environments. However, they rely on relatively stable environmental assumptions. In dynamic scenarios, continuous target movement can rapidly invalidate precomputed partitions and assignments, while the stage-wise structure limits fast global reallocation. Moreover, sparse and skewed detection events may propagate outdated information across stages, reducing adaptability in dynamic cooperative search.

B. Learning-Based Methods

MARL has emerged as a promising paradigm for multi-UAV collaboration, particularly under the centralized training with decentralized execution (CTDE) framework [29]. Value-decomposition methods such as QMIX [30] enable scalable coordination by factorizing joint value functions, while COMA [31] addresses credit assignment through counterfactual reasoning. Hierarchical MARL [32], [33] further introduces multi-level policy structures to improve coordination efficiency. Coach–Player frameworks (COPA) provide high-level guidance or abstract goals to lower-level agents, facilitating structured cooperation and improved scalability.

Despite these advances, two critical challenges remain in dynamic search tasks. First, centralized or hierarchical coordination often assumes reliable global information aggregation, which may be impractical under communication constraints or sensing delays. Second, search problems typically exhibit sparse and heavy-tailed reward distributions, where rare detection events generate large reward spikes. Such distributions increase variance in advantage estimation and destabilize conventional expectation-based updates. Existing scalable or hierarchical MARL frameworks primarily focus on policy decomposition or agent scalability, but do not explicitly model temporal belief evolution nor incorporate robustness mechanisms tailored to heavy-tailed rewards.

In contrast, our framework integrates hierarchical coordination with temporally evolving spatial belief modeling and heavy-tail-aware optimization, enabling both adaptive search reallocation and stable multi-agent learning in dynamic environments.

III. PROBLEM FORMULATION

A. Hierarchical System Design

We consider a heterogeneous multi-UAV system deployed in a bounded 3D mission space of dimensions $H \times W$ for cooperative dynamic target search. Due to altitude-dependent sensing resolution and communication range, UAVs operating at different heights exhibit asymmetric perception and connectivity capabilities. This inherent information asymmetry motivates a two-tier hierarchical architecture that balances global situational awareness and local detection precision.

As illustrated in Fig. 2, the system consists of two layers:

- **High-altitude UAVs (Hubs):** Operating at higher altitudes, hubs possess long-range sensing and broader communication coverage, enabling wide-area monitoring and coordination across multiple agents. However, their sensing resolution degrades with altitude, leading to increased false positives and missed detections for small or fast-moving targets. Rather than directly participating in motion planning, hubs act as information aggregators. Each hub collects local maps from nearby low-altitude UAVs, fuses them into a spatial suspicion heatmap representing regional belief distribution, and periodically broadcasts this fused map to guide agents. Importantly, hubs do not directly output control actions, avoiding centralized decision bottlenecks.
- **Low-altitude UAVs (Agents):** Agents operate at lower altitudes with limited sensing range but higher resolution, making them suitable for accurate target verification. Each agent follows a decentralized policy based on its local observation and the periodically updated fused heatmap. Specifically, agents upload local maps to the hub, receive aggregated belief information, and integrate it into their decision-making process. This bidirectional information exchange allows agents to leverage both precise local sensing and broader contextual awareness without requiring full-network synchronization.

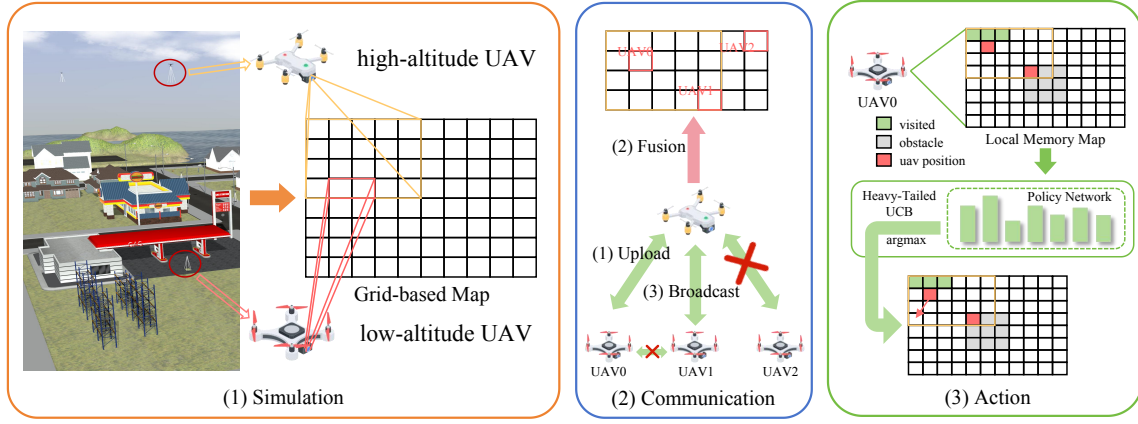


Fig. 2. Overview of the multi-UAV search system for dynamic targets in a bounded mission area. The system consists of two hierarchical UAV layers: high-altitude UAVs aggregate and broadcast information for global suspicion fusion, and low-altitude UAVs perform localized search tasks.

This altitude-aware division of labor addresses sensing asymmetry and communication constraints, enabling scalable coordination while preserving decentralized execution.

B. Scenario-Specific Challenges

Dynamic target search differs fundamentally from conventional coverage tasks in both perception reliability and reward structure.

First, sensing quality is altitude-dependent. Although high-altitude UAVs provide broad coverage, their degraded resolution introduces substantial detection uncertainty, including frequent false positives and missed detections. Consequently, the global belief representation may contain noisy or outdated information, requiring robust fusion mechanisms.

Second, dynamic search inherently produces sparse and highly skewed reward signals. Meaningful rewards are generated only when a UAV successfully detects a target at the correct location and time. Most time steps yield zero reward, while rare detection events produce disproportionately large reward spikes. Such behavior naturally induces heavy-tailed reward distributions.

Formally, let R_{target} denote the reward upon successful detection. We model R_{target} as a heavy-tailed random variable satisfying:

$$\mathbb{E} [|R_{\text{target}}|^{1+\epsilon}] < \infty, \quad \epsilon \in (0, 1] \quad (1)$$

which implies finite $(1 + \epsilon)$ -moment but potentially unbounded or extremely large variance. Smaller ϵ corresponds to heavier tails, reflecting the presence of rare yet high-magnitude rewards. Under such conditions, conventional mean-based estimators may suffer from high variance and unstable policy updates.

In summary, effective learning in dynamic multi-UAV search requires simultaneously addressing (1) altitude-induced sensing uncertainty and belief evolution, and (2) sparse, heavy-tailed reward structures that challenge standard reinforcement learning estimators.

IV. METHODOLOGY

A. Dec-POMDP Formulation

We formulate the cooperative multi-UAV search task as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP), defined as $\mathcal{M} = \langle \mathcal{N}, \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{O}, \mathcal{Z}, \mathcal{R}, \gamma \rangle$. Here, $\mathcal{N} = \{1, \dots, N\}$ denotes the set of N low-altitude UAV agents. The global state space $\mathcal{S}$ includes agent positions, obstacle configurations, target states, and the global suspicion heatmap. Each agent receives a local observation $o_i \in \mathcal{O}_i$ represented as an $H \times W \times 4$ tensor containing: (1) target distribution estimates, (2) suspicion heatmap values, (3) teammate positions, and (4) visited grid indicators.

Agents select discrete motion actions $a_i \in \mathcal{A}_i$, consisting of 25 neighboring movements within a 5×5 grid plus a stationary action. All candidate actions are pre-validated to ensure collision-free transitions. The joint action $\mathbf{a} = (a_1, \dots, a_N)$ drives state transitions via $\mathcal{P}$. The reward function $\mathcal{R}$ is sparse and asymmetric. A large positive reward R_{target} is granted upon successful detection, while a significantly smaller shaping reward encourages exploration of previously unvisited regions. Since $R_{\text{target}} \gg R_{\text{coverage}}$, the resulting reward distribution is inherently heavy-tailed, characterized by rare but high-magnitude events. Each agent maximizes the expected discounted return with discount factor γ . Episodes terminate when all targets are detected or a maximum horizon T_{max} is reached.

B. Suspicion Heatmap as Temporal Belief Memory

To address detection uncertainty and dynamic target motion, we introduce a suspicion heatmap as a global temporal belief representation. Unlike instantaneous detections, the heatmap integrates probabilistic evidence over time, allowing the system to accumulate weak signals while discounting outdated information.

Each grid cell maintains a belief value $s \in [0, 1]$. When a high-altitude UAV reports a detection with confidence score strength, the belief is updated as:

$$s \leftarrow 1 - (1 - s)(1 - g), \quad g = \min(1, \lambda \cdot \text{strength}) \quad (2)$$

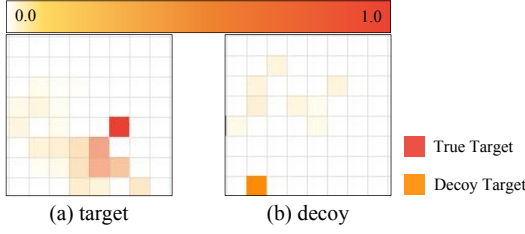


Fig. 3. Suspicion heatmap evolution. (a) Repeated, possibly low-confidence detections progressively reinforce the true-target belief. (b) Fewer decoy detections produce lower confidence that gradually decays without further evidence.

which corresponds to multiplicative fusion of independent detection evidence. Repeated detections reinforce belief while preserving boundedness.

In the absence of new evidence, belief decays exponentially:

$$s \leftarrow s \cdot \delta, \quad \delta \in (0, 1) \quad (3)$$

This decay mechanism enables adaptive forgetting, preventing stale information from dominating decision-making in dynamic environments.

The resulting heatmap serves as a temporally evolving spatial prior, amplifying consistent signals and suppressing noise. It provides structured guidance for decentralized agents without requiring centralized control. Fig. 3 illustrates the evolution of the heatmap under repeated detections and decoy observations.

C. Heavy-Tailed UCB for Robust Exploration

Conventional intrinsic exploration methods, such as entropy regularization or count-based bonuses, often exhibit high variance or poor scalability in large-scale multi-agent settings. Upper Confidence Bound (UCB) strategies provide a principled alternative by balancing exploration and exploitation through confidence intervals. However, classical UCB methods rely on sub-Gaussian noise assumptions [34]. In dynamic search tasks with heavy-tailed rewards, variance may be extremely large or even undefined under finite samples, invalidating standard concentration bounds.

To address this, we adopt a heavy-tailed UCB formulation derived from concentration inequalities that require only finite $(1 + \epsilon)$ -moment assumptions [35]. The exploration index for region i at time t is defined as

$$U_{t,i} = f(x_{t,i}; \theta_{t-1}) + \gamma_t^{\text{heavy}} \sqrt{g_{t,i}^\top Z_{t-1}^{-1} g_{t,i}} \quad (4)$$

where $x_{t,i}$ denotes the contextual feature of region i at step t , and $f(x_{t,i}; \theta)$ is a neural network estimating the expected reward. The feature representation is given by $g_{t,i} = \frac{1}{\sqrt{m}} \nabla_\theta f(x_{t,i}; \theta_{t-1})$. The design matrix is defined as

$$Z_t = \lambda I + \sum_{s=1}^t g_{s,i_s} g_{s,i_s}^\top. \quad (5)$$

The confidence width γ_t^{heavy} is

$$\gamma_t^{\text{heavy}} = C \cdot M_\epsilon^{\frac{1}{1+\epsilon}} (\log \det Z_{t-1})^{\frac{\epsilon}{1+\epsilon}} t^{\frac{1-\epsilon}{2(1+\epsilon)}}, \quad (6)$$

where $\epsilon \in (0, 1]$ controls the heaviness of the reward tail, M_ϵ bounds the $(1 + \epsilon)$ -moment of the noise, and C is a constant. This confidence width remains valid under heavy-tailed noise, ensuring reliable exploration even when extreme rewards occur.

D. Robust MAPPO with Median-of-Means

Heavy-tailed rewards significantly affect policy optimization. Standard PPO relies on mean-based return and advantage estimation, which are highly sensitive to extreme trajectories. A small number of large rewards may dominate gradient updates, leading to unstable learning dynamics.

To mitigate this effect, we replace mean-based aggregation with the Median-of-Means (MoM) estimator. Given n samples partitioned into k blocks, MoM computes the mean within each block and then takes the median across block means. Under finite $(1 + \epsilon)$ -moment assumptions, MoM provides sub-Gaussian-type concentration guarantees even when variance is large.

The MoM reward is defined as:

$$\tilde{r}_t = \text{MoM}(\{r_t^{(j)}\}_{j=1}^k) \quad (7)$$

The block number k controls the robustness–variance trade-off: larger k increases robustness to outliers but raises estimator variance, while smaller k reduces variance but weakens robustness.

Using $\tilde{r}_t$, we compute value targets and advantages:

$$y_t = \tilde{r}_t + \gamma V_\phi(s_{t+1}), \quad (8)$$

$$\delta_t = \tilde{r}_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (9)$$

$$A_t = \text{GAE}(\delta_t) \quad (10)$$

We further apply MoM aggregation to PPO losses:

$$\mathcal{L}^\pi = \text{MoM}(-\min(r_t A_t, \text{clip}(r_t, 1 \pm \epsilon) A_t)), \quad (11)$$

$$\mathcal{L}^V = \text{MoM}((V_\phi(s_t) - y_t)^2), \quad (12)$$

$$\mathcal{L}^H = \text{MoM}(\mathcal{H}(\pi_\theta(\cdot | o_t))). \quad (13)$$

The overall objective is:

$$\mathcal{L} = \mathcal{L}^\pi + c_v \mathcal{L}^V - c_{\text{ent}} \mathcal{L}^H \quad (14)$$

The coefficient c_{ent} controls the strength of entropy regularization, encouraging policy exploration during training.

V. EXPERIMENTS

A. Experimental Setup

1) *Simulation Environment*: All experiments are conducted in a high-fidelity 3D simulation environment built upon Gazebo, integrated with PX4 and ROS through the XTDrone platform [36]. This modular framework provides realistic UAV dynamics, sensor feedback, and low-level flight control, enabling reliable evaluation of multi-UAV coordination policies.

Dynamic Target Motion. All experimental results reported in this paper are obtained under dynamic target search settings. Targets move continuously throughout each

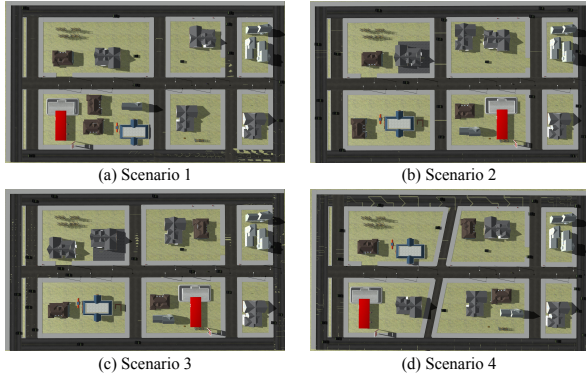


Fig. 4. Randomized urban simulation scenarios. Each panel (a-d) shows a different scenario with varied layouts of roads, buildings, and vehicles.

episode following stochastic motion policies constrained by road structures. At each time step, a target updates its position with velocity sampled from a predefined range and applies random directional perturbations to simulate realistic movement behavior. Consequently, previously observed high-likelihood regions may become obsolete over time, requiring adaptive belief updates and robust policy learning.

Environment Randomization. To evaluate generalization ability, we construct multiple randomized urban environments (Fig. 4), each containing procedurally generated layouts of roads, buildings, and vehicles. Obstacle distributions vary significantly across scenarios, resulting in different visibility constraints and navigation complexity. All quantitative comparisons are averaged across these distinct layouts. For decision-making, each environment is discretized into a grid map of size 168×92 . At each decision step, every low-altitude UAV selects a target grid cell as its navigation goal. The selected goal is passed to Ego-Planner v2 [37], which generates dynamically feasible and collision-free trajectories under UAV dynamic constraints and obstacle avoidance requirements.

To simulate perception uncertainty, we collect annotated simulation data to train a YOLOv3-based object detector. The detector provides probabilistic confidence scores used for updating the suspicion heatmap. Unless otherwise specified, detection results include both true positives and decoy-induced false detections to reflect realistic sensing noise.

2) Baselines: We compare our method with representative approaches from rule-based planning, flat MARL, and hierarchical MARL. **Rule-Based Planning Methods:** (1) *Iterative Utility-Based Method*, where each UAV greedily selects actions based on exploration gain and redundancy penalties derived from the shared suspicion map; (2) *Metaheuristic-Based Search*, which updates trajectories using swarm-inspired optimization guided by aggregated map information; (3) *TSP-Based Method*, where high-suspicion regions are allocated via short-horizon task assignment solved through localized TSP planning. **Flat MARL Methods:** Representative CTDE-based approaches including QMIX [30], COMA [31], and AM-MAPPO [29], [38]. **COPA:** COPA [33], a coach-player hierarchical MARL framework in which a high-level

coach policy provides abstract goals to decentralized low-level agents to improve coordination. All learning-based baselines are trained under identical dynamic target settings with the same observation space and action discretization.

3) Evaluation Metrics: We adopt the following quantitative metrics: (1) **Target Coverage (Tgt):** The total number of true targets successfully detected before episode termination. (2) **Overlap Ratio (OR):** The percentage of grid cells visited by more than one UAV, reflecting search redundancy. (3) **Search Time (Time):** The number of steps required to detect all targets or reach the maximum episode length.

4) Implementation Details: All low-altitude UAVs share policy parameters. Both actor and critic networks are single-layer MLPs with hidden dimension 64 and ReLU activation. Training hyperparameters are set as follows: buffer size 3000, learning rate 7×10^{-4} , discount factor $\gamma = 0.95$, clipping ratio 0.2, entropy coefficient 0.01, MoM block number $k = 8$ and value loss coefficient 0.5. We apply value normalization, advantage normalization, and generalized advantage estimation ($\lambda = 0.95$). Gradients are clipped with a maximum norm of 10.0. All reported results are averaged over 10 independent runs with different random seeds, and standard deviations are reported where applicable.

B. Main Results

Table I reports the performance under dynamic target search across four randomized urban layouts with distinct obstacle distributions. All methods are evaluated under identical target motion and sensing uncertainty. Our method consistently achieves the highest target coverage in all scenarios (6.00 targets) while maintaining the lowest overlap ratio and significantly shorter search time (e.g., 117.0 s in Scenario 1). In contrast, the best rule-based baseline (TSP-Based) discovers at most 5.70 targets and requires over 178 s with substantially higher overlap.

Rule-based planners (Iterative, Meta-heuristic, TSP-Based) rely on short-horizon heuristics and lack temporal belief modeling. As targets move and detections become noisy, previously highlighted regions become outdated, leading to repeated revisits and inefficient search. Flat MARL methods (QMIX, COMA, AM-MAPPO) perform even worse, discovering only 1.40–2.30 targets in most scenarios due to unstable learning under sparse and heavy-tailed rewards. COPA improves coordination compared to flat MARL but remains sensitive to noisy detections because it lacks temporally evolving belief modeling and robust reward aggregation.

The superior performance of our framework arises from the synergy of three mechanisms. First, the suspicion heatmap with exponential decay enables adaptive belief evolution, allowing agents to shift search focus as targets move. Second, heavy-tailed UCB exploration prioritizes uncertain regions without overreacting to extreme reward samples. Third, MoM-based policy optimization stabilizes training under heavy-tailed returns, preventing rare detection events from dominating gradient updates. Together, these components enable efficient, low-redundancy coordination and robust convergence in dynamic multi-UAV search tasks.

TABLE I
DYNAMIC-SEARCH METHOD COMPARISON ACROSS FOUR RANDOMIZED URBAN LAYOUTS WITH DISTINCT OBSTACLES.

Methods	Scenario 1			Scenario 2			Scenario 3			Scenario 4		
	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$
Iterative	5.20	0.4832	182.6	5.10	0.4049	198.5	4.90	0.4467	198.3	4.90	0.3644	200.0
Meta-heuristic	4.80	0.4197	200.0	4.90	0.3837	200.0	4.60	0.4389	200.0	4.40	0.3742	200.0
TSP-Based	5.40	0.4460	188.6	5.70	0.4060	178.6	5.30	0.3868	192.5	5.10	0.3997	196.8
QMIX	1.50	0.1028	200.0	1.80	0.1227	200.0	1.50	0.1278	200.0	1.40	0.1028	200.0
COMA	2.20	0.1432	200.0	2.30	0.1388	200.0	2.10	0.1804	200.0	1.90	0.1136	200.0
AM-MAPPO	1.60	0.3340	200.0	1.80	0.3654	200.0	1.40	0.3315	200.0	1.60	0.3165	200.0
COPA	5.10	0.5302	171.5	4.90	0.5237	171.4	4.80	0.5190	173.8	4.70	0.4993	192.7
Ours	6.00	0.0934	117.0	6.00	0.0902	127.6	6.00	0.0836	124.3	6.00	0.0900	120.9

Note: Tgt = Target coverage (%); OR = Overlap ratio (%); Time = Search time (s). Arrows ($\uparrow/\downarrow$) indicate optimization direction.

TABLE II
ABLATION STUDY ON KEY COMPONENTS.

Methods	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$
Full Model (Ours)	6.00	0.090	120.9
- w/o Suspicion Heatmap	4.80	0.245	181.2
- w/o Heavy-Tailed UCB	1.80	0.328	200.0
- w/o MoM Rewards	1.60	0.332	200.0

TABLE III
IMPACT OF MoM BLOCK NUMBER k ON FINAL PERFORMANCE.

k	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$
1 (Mean)	1.60	0.3320	200.0
4	5.50	0.1312	146.0
8	6.00	0.0934	117.0
16	5.80	0.0525	134.0

C. Ablation Studies

1) *Effectiveness of Key Components:* We further analyze the role of key components in our framework through ablation studies. As shown in Table II, each module contributes distinctly to overall performance. To assess the impact of the suspicion heatmap, we replace it with raw single-step detection results. This prevents UAVs from accumulating high-level guidance over time, leading to less informed decision-making and significantly longer search times. In contrast, the Heavy-Tailed UCB and MoM Rewards are essential for addressing the inherently heavy-tailed reward distribution. Removing either results in unstable learning dynamics, where the final policy fails to converge and behaves similarly to random search, with poor target coverage and high redundancy. Overall, the full model consistently achieves the best performance across all metrics. These results confirm that the suspicion heatmap enhances search efficiency via temporal fusion of detections, while Heavy-Tailed UCB and MoM Rewards are indispensable for stable and robust training under challenging reward conditions.

2) *Impact of MoM Block Size:* Table III shows the impact of the MoM block number k . When $k = 1$, the estimator reduces to the standard mean update, which is highly sensitive to heavy-tailed rewards and leads to unstable learning, discovering only 1.60 targets on average. Increasing k significantly improves performance: with $k = 4$ the policy recovers,

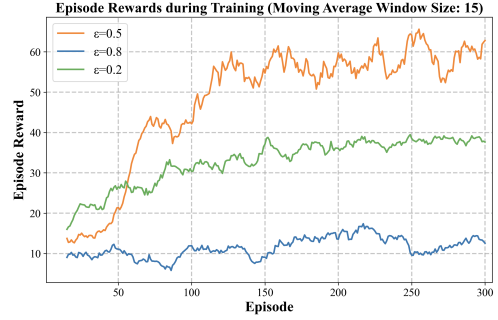


Fig. 5. Episode rewards during training with varying values of the exploration parameter ϵ .

and $k = 8$ achieves the best overall results, discovering all targets with the shortest search time (117.0). When $k = 16$, the overlap ratio further decreases but the search time slightly increases, indicating that overly large block numbers may introduce additional estimation noise. Overall, moderate block sizes provide the best balance between robustness and learning efficiency.

3) *Sensitivity to ϵ in Exploration:* In our Heavy-Tailed UCB design, the parameter $\epsilon \in (0, 1]$ characterizes the heaviness of the reward distribution tail and shapes the exploration confidence bound. Smaller ϵ values produce wider bounds, leading to conservative but stable exploration, while larger values shrink the bounds faster, risking premature exploitation under noisy signals. As shown in Fig. 5, when ϵ is set smaller than the true value, convergence remains theoretically valid but proceeds slowly due to excessive exploration. In contrast, larger values invalidate the heavy-tailed assumptions, causing unstable learning. The intermediate choice $\epsilon = 0.5$ achieves the best balance, showing fast and stable convergence with low variance across runs.

D. Robustness and Scalability

1) *Robustness to Sensing Uncertainty:* To assess robustness under different detection capabilities, we simulate high-level UAV sensing probabilities instead of using a trained YOLO detector. Specifically, we vary the true detection probability p_{true} and the false-alarm probability p_{decoy} . Table IV reports performance under three representative settings. The results show that our method reliably discovers nearly all

TABLE IV
PERFORMANCE UNDER DIFFERENT SENSING PROBABILITIES

Sensing Probability	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$
$p_{\text{true}} = 0.5, p_{\text{decoy}} = 0.5$	5.80	0.1520	155.30
$p_{\text{true}} = 0.7, p_{\text{decoy}} = 0.3$	5.93	0.1425	138.60
$p_{\text{true}} = 0.9, p_{\text{decoy}} = 0.1$	6.00	0.1380	122.40

Note: Detection confidence sampling intervals: true targets [0.35, 0.60], decoy targets [0.20, 0.45].

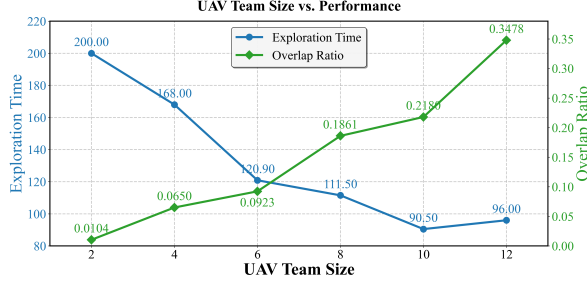


Fig. 6. Impact of UAV team size with six targets: search time (left axis) and overlap ratio (right axis). For two UAVs, 200 steps indicates that not all targets were found before the episode limit.

targets regardless of sensing quality. Even in the extreme case of $p_{\text{true}} = 0.5$ and $p_{\text{decoy}} = 0.5$, the discovery rate remains high and search time increases slightly. As sensing quality improves, the overlap ratio decreases, and the search process becomes faster. These findings show that the proposed suspicion heatmap robustly integrates uncertain, noisy detections from high-altitude UAVs to support reliable decision-making in hierarchical multi-UAV search.

2) *Impact of UAV Team Size:* We examine how the number of low-level UAVs affects search performance by varying team sizes from 2 to 12, while keeping the number of targets fixed at six. As shown in Fig. 6, increasing the team size generally speeds up the search: completion time drops from 168 steps with 4 UAVs to 90.5 steps with 10 UAVs. However, with only 2 UAVs, not all targets are found within the 200-step episode limit, indicating insufficient coverage. Conversely, expanding the team to 12 UAVs brings minimal time gains but significantly increases the overlap ratio, reflecting redundant search. These results highlight a clear trade-off: a moderate team size (around 6 to 10 UAVs) balances speed and redundancy, while too few UAVs hinder coverage and too many lead to inefficient coordination.

3) *Adaptability to Varying Numbers of Targets:* To evaluate the adaptability of our method under different task densities, we vary the number of targets in the environment from 2 to 10 while keeping all other settings fixed. All low-altitude UAVs share the same distributed policy network, which is trained under the configuration of six UAVs. As shown in Table V, our method consistently discovers nearly all targets across all settings, even as the number of targets increases. Importantly, the overlap ratio remains low and stable, indicating effective coordination among UAVs without excessive redundancy. Although completion time naturally increases with more targets, the growth is modest and

TABLE V
PERFORMANCE COMPARISON UNDER DIFFERENT NUMBERS OF TARGETS

Number of Targets	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$
2	2.00	0.1402	92.30
4	4.00	0.0997	100.30
6	6.00	0.0900	120.90
8	8.00	0.1220	125.80
10	9.67	0.1440	140.40

Note: Results are obtained using a shared policy network trained with six low-altitude UAVs.

TABLE VI
IMPACT OF GRID SIZE ON DYNAMIC SEARCH PERFORMANCE.

Grid Size	Tgt $\uparrow$	OR $\downarrow$	Time $\downarrow$
84×46	6.00	0.1013	73.0
168×92	6.00	0.0934	117.0
336×184	5.80	0.0200	186.6

scales smoothly, demonstrating that our method maintains high search efficiency under denser task distributions. These results confirm the strong generalization capability of our framework to environments with varying target densities.

E. Impact of Grid Size

We evaluate the impact of map range by varying the grid size $H \times W$ while keeping the physical scale per grid cell fixed. Changing $H \times W$ therefore modifies the overall mission area and task difficulty under continuous target motion. Table VI reports the results. When the map range is small, the search space is limited and the system completes the task quickly. As the map size increases to 168×92 , the search time grows moderately while maintaining full target discovery. In the largest environment, the search time further increases due to the enlarged exploration space, while the overlap ratio decreases, indicating less redundant coverage among agents. Overall, the proposed framework maintains strong performance across different map ranges, suggesting good generalization to varying search space scales.

VI. CONCLUSIONS

In this work, we introduce a hierarchical MARL framework for decentralized UAV swarm search in large-scale dynamic environments with sparse and heavy-tailed reward distributions. Our approach leverages altitude-based heterogeneity: high-altitude UAVs generate noisy but wide-range observations to update a shared suspicion heatmap, while low-altitude UAVs use this memory to guide fine-grained search while maintaining autonomy. By integrating a heavy-tailed UCB exploration mechanism and a Median-of-Means-based MAPPO algorithm, our method achieves robust credit assignment under extreme sparse reward conditions. Evaluations in a 3D simulation environment demonstrate that our approach outperforms several strong baselines in terms of detection rate, efficiency, and noise resistance, without requiring centralized control or dense communication, making it feasible for deployment in practical scenarios with limited bandwidth and sensing resources.

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